

Paper 1 Literature Survey

Title: *A Systematic Literature Study of Machine Learning Techniques Based Intrusion Detection: Datasets, Models, Challenges, and Future Directions*

Objective

- To review machine learning techniques used in **Intrusion Detection Systems (IDS)**.

Method / Workflow

- PRISMA-based systematic literature review.
- Research papers collected from IEEE, ACM, Springer, ScienceDirect, and Google Scholar.
- **1598 papers identified → 104 papers selected for analysis.**

Techniques Covered

- Supervised learning: Decision Tree, SVM, Random Forest
- Unsupervised learning methods
- Deep learning models: CNN, RNN, LSTM
- Hybrid IDS approaches.

Datasets Discussed

- NSL-KDD
- CICIDS2017
- UNSW-NB15

Key Findings

- Machine learning improves detection of cyber attacks.
- Deep learning models achieve higher detection accuracy.

Limitations

- Many datasets are outdated.
- Dataset imbalance affects IDS performance.
- Difficulty detecting zero-day attacks.

Contribution / Difference

- This paper **does not propose a new IDS model**.
- It provides a **comprehensive survey of ML-based IDS research and datasets**.

Paper 2 Literature Survey

Title: AI-Powered Intrusion Detection Systems: Enhancing Real-Time Network Threat Monitoring

Objective

- To analyze the use of **Artificial Intelligence for real-time intrusion detection**.

Method / Workflow

- Systematic review of IDS research from **2015–2024**.
- **1480 papers collected** → **28 papers analyzed**.

Techniques Used

- Machine learning: SVM, Random Forest, Decision Trees
- Deep learning: CNN, LSTM
- Hybrid AI detection models.

Datasets Used

- KDD Cup 99
- NSL-KDD
- UNSW-NB15
- CICIDS2017

Key Findings

- AI-based IDS improves **real-time threat detection**.
- Deep learning models achieve **95–98% detection accuracy**.

Limitations

- High computational cost of deep learning.
- Dataset imbalance issues.
- Vulnerability to adversarial attacks.

Contribution / Difference

- Focuses specifically on **AI-driven IDS for real-time network monitoring** rather than general IDS surveys.

Paper 3 Literature Survey

Title: *Robust Machine Learning Based Intrusion Detection System Using Simple Statistical Techniques in Feature Selection*

Objective

- To develop a **lightweight IDS using statistical feature selection methods**.

Method / Workflow

1. Feature selection using statistical techniques:
 - Pearson correlation
 - Chi-square test
 - ANOVA
2. Selected features used for ML classification.

Techniques Used

- Random Forest
- AdaBoost
- Decision Tree classifiers.

Datasets Used

- IoTID20 dataset
- NSL-KDD dataset.

Key Findings

- Feature selection reduces computational cost.
- Proposed model achieved **~99.9% accuracy**.

Limitations

- Model performance depends on dataset quality.

- More improvement possible using deep learning methods.

Contribution / Difference

- Unlike survey papers, this paper **proposes an actual IDS model**.
 - Focuses on **feature selection to improve IDS performance**.
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Paper 4 Literature Survey

Title: *Intrusion Detection Systems in IoT Based on Machine Learning: A State of the Art*

Objective

- To review machine learning techniques used for **intrusion detection in IoT environments**.

Method / Workflow

- Analysis of recent IDS research related to IoT security.
- Comparison of ML models based on detection rate and false positives.

Techniques Covered

- Machine learning algorithms
- Deep learning methods
- Feature selection and anomaly detection approaches.

Datasets Discussed

- CICIDS-2017 dataset
- UNSW-NB15 dataset and other IoT datasets.

Key Findings

- ML-based IDS improves detection of unknown attacks in IoT networks.
- Anomaly detection techniques help identify new threats.

Limitations

- IoT devices have limited computational resources.
- Higher false positives may occur in anomaly-based IDS.

Contribution / Difference

- Focuses specifically on **IoT network security and IDS applications in IoT systems**.
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Short Comparison of All Papers

- **Paper 1:** Provides a **comprehensive survey of ML-based IDS techniques and datasets**.
- **Paper 2:** Focuses on **AI-based IDS for real-time network threat detection**.
- **Paper 3:** Proposes a **lightweight IDS model using statistical feature selection**.
- **Paper 4:** Reviews **machine learning intrusion detection methods specifically for IoT environments**.

Paper 5 Literature Survey

Title: *An Autonomous Intrusion Detection System Using an Ensemble of Advanced Learners*

Objective

- To develop an **autonomous intrusion detection system** capable of detecting **zero-day attacks with minimal human intervention**.

Method / Workflow

1. Network traffic features extracted using IDS tools.
2. Three learning models trained in parallel.
3. Results combined using **majority voting and OR logic**.

Techniques Used

- Convolutional Neural Network (CNN)
- Gated Recurrent Unit (GRU)
- Random Forest.

Dataset Used

- NSL-KDD dataset.

Key Findings

- Achieved **87.28% accuracy on KDDTest+** dataset.
- Ensemble learning improved detection capability.

Limitations

- Accuracy lower for difficult datasets.
- Computational complexity due to multiple models.

Contribution / Difference

- Uses **ensemble deep learning models together** for intrusion detection.
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Paper 6 Literature Survey

Title: *AutoIDS: Autoencoder Based Intrusion Detection System*

Objective

- To design a **semi-supervised anomaly detection IDS** using autoencoders.

Method / Workflow

1. Train autoencoder using **normal network traffic only**.
2. Detect anomalies using **reconstruction error and sparse representation**.
3. Cascade architecture with two detectors.

Techniques Used

- Sparse Autoencoder
- Encoder-Decoder Neural Networks.

Dataset Used

- NSL-KDD dataset.

Key Findings

- Achieved **90.17% detection accuracy**.
- Cascade approach reduces computation cost.

Limitations

- Performance depends on threshold selection.
- Limited evaluation on real-world datasets.

Contribution / Difference

- Uses **semi-supervised deep learning without attack labels**.
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Paper 7 Literature Survey

Title: *Adaptive Intrusion Detection System Leveraging Dynamic Neural Models with Adversarial Learning for 5G/6G Networks*

Objective

- To design a **robust IDS for 5G/6G networks** using adversarial learning and dynamic neural networks.

Method / Workflow

1. Extract network traffic features.
2. Train dynamic neural network classifier.
3. Apply adversarial learning to improve robustness.

Techniques Used

- Dynamic Neural Networks
- Adversarial Machine Learning
- Incremental Learning.

Dataset Used

- NSL-KDD dataset.

Key Findings

- Achieved **82.33% accuracy for multiclass attack detection**.
- Incremental learning reduces retraining cost.

Limitations

- Lower accuracy compared to deep learning IDS.

- Needs testing on real 5G datasets.

Contribution / Difference

- Focuses specifically on **security challenges in 5G/6G networks**.
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Paper 8 Literature Survey

Title: *Machine Learning-Based Intrusion Detection Systems: Capabilities, Methodologies, and Open Research Challenges*

Objective

- To analyze **capabilities and future research directions of ML-based IDS systems**.

Method / Workflow

- Comprehensive analysis of ML techniques used in intrusion detection.
- Evaluation of IDS workflow including preprocessing, training, and inference.

Techniques Covered

- Traditional ML: SVM, KNN, Decision Tree
- Deep Learning: CNN, RNN, Autoencoders
- Advanced methods: Graph Neural Networks (GNN), Large Language Models (LLM).

Key Findings

- ML improves detection of **zero-day attacks**.
- ML can detect **intrusions in encrypted traffic**.

Limitations

- ML models require large datasets.
- Explainability and trust issues remain.

Contribution / Difference

- Discusses **future IDS technologies such as GNNs and LLMs**.
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Paper 9 Literature Survey

Title: *Advancements in Machine Learning-Based Intrusion Detection in IoT: Research Trends and Challenges*

Objective

- To review machine learning techniques used for **detecting DDoS attacks in IoT networks**.

Method / Workflow

- Systematic literature review using **PRISMA methodology**.
- Analysis of datasets, ML models, and deployment environments.

Techniques Used

- Decision Tree
- Random Forest
- Deep learning models.

Datasets Used

- BoT-IoT dataset
- TON_IoT dataset.

Key Findings

- Random Forest and Decision Tree achieve **>99% accuracy in many cases**.
- Lightweight ML models preferred for IoT.

Limitations

- IoT devices have limited computing resources.
- Need larger datasets for better IDS models.

Contribution / Difference

- Focuses specifically on **IoT-based DDoS attack detection trends**.
-

Paper 10 Literature Survey

Title: *Deep Learning for Intrusion Detection in Emerging Technologies: A Comprehensive Survey*

Objective

- To analyze **deep learning approaches for intrusion detection in emerging technologies**.

Method / Workflow

- Literature review of IDS methods applied in:
 - Cloud computing
 - IoT
 - Edge computing
 - Industrial control systems.

Techniques Covered

- Deep Neural Networks (DNN)
- CNN
- RNN
- GAN
- Transformers.

Key Findings

- Deep learning models achieve higher detection accuracy than traditional ML.
- Useful for detecting complex cyber attacks.

Limitations

- High false positive rates.
- Difficult deployment in real systems.

Contribution / Difference

- Focuses on **deep learning IDS for emerging technologies**.
-

Paper 11 Literature Survey

Title: *Intrusion Detection System Based on Machine Learning Using Least Square Support Vector Machine*

Objective

- To design a **high-accuracy intrusion detection system using LS-SVM classifier**.

Method / Workflow

1. Feature selection using exhaustive feature selection algorithm.
2. Train classifier using selected features.
3. Evaluate performance on multiple datasets.

Techniques Used

- Least Square Support Vector Machine (LS-SVM)
- Feature Selection
- Ensemble classifiers.

Datasets Used

- NSL-KDD
- CICIDS2017
- UNSW-NB15.

Key Findings

- Accuracy achieved:
 - **99.3% (NSL-KDD)**
 - **99.5% (CICIDS2017)**
 - **93.3% (UNSW-NB15).**

Limitations

- Feature selection increases preprocessing complexity.
- Model performance depends on dataset quality.

Contribution / Difference

- Combines **exhaustive feature selection with LS-SVM** to reduce false positives.
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Short Comparison of These Papers

- **Paper 5:** Ensemble deep learning IDS model.
- **Paper 6:** Autoencoder-based anomaly detection IDS.
- **Paper 7:** IDS designed for **5G/6G networks**.
- **Paper 8:** Comprehensive review of **ML-based IDS techniques and challenges**.
- **Paper 9:** Survey focusing on **IoT DDoS detection methods**.
- **Paper 10:** Review of **deep learning IDS for emerging technologies**.
- **Paper 11:** High-accuracy **LS-SVM-based intrusion detection model**.

Below is the **same format concise literature survey** for the **4 new papers** you uploaded. I kept it **short, structured, and line-by-line** so you can directly copy into your literature survey document.

Paper 12 Literature Survey

Title: *Survey of Machine Learning Based Intrusion Detection Methods for Internet of Medical Things*

Objective

- To analyze **machine learning-based intrusion detection techniques** for **Internet of Medical Things (IoMT)** systems.

Method / Workflow

1. Analyze IoMT architecture and security requirements.
2. Identify threats affecting IoMT systems.
3. Review machine learning IDS approaches at different IoMT layers.

IoMT Architecture

- Data acquisition layer (sensors and medical devices).
- Personal server layer.
- Medical server layer.

Techniques Covered

- Machine Learning algorithms such as:
 - SVM
 - Decision Tree
 - Random Forest
 - Neural Networks
- Deep Learning techniques.

Key Findings

- ML-based IDS can detect **zero-day attacks and abnormal behaviors**.
- Large IoMT data volumes help train accurate models.

Limitations

- IoMT devices have **limited computation and energy resources**.
- High data sensitivity raises privacy concerns.

Contribution / Difference

- Provides **detailed analysis of IDS solutions for all three IoMT layers** (device, transmission, and server levels).
-

Paper 13 Literature Survey

Title: *Survey of Network Intrusion Detection Methods from the Perspective of the Knowledge Discovery in Databases Process*

Objective

- To review network intrusion detection methods using the **Knowledge Discovery in Databases (KDD) process framework**.

Method / Workflow

The IDS process is divided into **five KDD stages**:

1. Data collection (network traffic capture).
2. Data preprocessing.
3. Data reduction (feature selection).
4. Data mining (ML algorithms).
5. Evaluation and interpretation.

Techniques Covered

- Data mining algorithms for intrusion detection:
 - Clustering
 - Classification
 - Anomaly detection models.

Key Findings

- Intrusion detection depends not only on algorithms but also on **data collection and preprocessing quality**.

Limitations

- Many IDS studies focus only on algorithms and ignore **data preparation and evaluation steps**.

Contribution / Difference

- Introduces a **KDD-based framework to analyze IDS research**, covering the entire detection pipeline.
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Paper 14 Literature Survey

Title: *Evaluation of Machine Learning Algorithms for Intrusion Detection System*

Objective

- To evaluate the performance of **different machine learning algorithms for intrusion detection**.

Method / Workflow

1. Use **KDD intrusion detection dataset** for experiments.
2. Train multiple machine learning classifiers.
3. Evaluate performance using accuracy and error metrics.

Algorithms Tested

- J48 Decision Tree
- Random Forest

- Random Tree
- Decision Table
- Naïve Bayes
- Bayes Network
- Multi-Layer Perceptron.

Key Findings

- Random Forest achieved the **highest accuracy** among tested algorithms.
- Decision Table produced the **lowest false negative rate**.

Limitations

- Dataset imbalance affects IDS performance.
- Single ML models may not detect all attack types effectively.

Contribution / Difference

- Provides **experimental comparison of several ML classifiers for IDS performance evaluation**.
-

Paper 15 Literature Survey

Title: *Building an Efficient Intrusion Detection System Based on Feature Selection and Ensemble Classifier*

Objective

- To design an **efficient intrusion detection system using feature selection and ensemble learning**.

Method / Workflow

1. Data preprocessing of IDS datasets.
2. Feature selection using **CFS-BA algorithm**.
3. Train multiple classifiers.
4. Combine classifiers using ensemble voting technique.

Techniques Used

- Feature selection: Correlation-based Feature Selection (CFS).

- Ensemble classifiers:
 - C4.5 Decision Tree
 - Random Forest
 - Forest by Penalizing Attributes.

Datasets Used

- NSL-KDD
- AWID
- CIC-IDS2017.

Key Findings

- Ensemble classifier improves **attack detection rate and classification accuracy**.
- Feature selection reduces dimensionality and computational complexity.

Limitations

- Ensemble models increase computational cost.
- Performance depends on dataset quality.

Contribution / Difference

- Combines **feature selection and ensemble learning** to improve IDS accuracy and stability.
-

Short Comparison of These Papers

- **Paper 12:** Survey of ML-based IDS for **IoMT healthcare systems**.
- **Paper 13:** IDS analysis using **KDD data mining process framework**.
- **Paper 14:** Experimental comparison of **multiple ML algorithms for IDS**.
- **Paper 15:** Proposed IDS using **feature selection + ensemble classifiers**.